

SHASHANK AGRAWAL

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CAREER OBJECTIVE

A motivated and growth-oriented individual aiming to build a career in the Artificial Intelligence and Machine Learning domain at an innovative organization where research and real-world applications intersect. Possesses strong problem-solving abilities, proficiency in working with AI tools, and a continuous learning mindset, enabling effective contributions to collaborative teams. Seeks opportunities for mentorship, professional development, and hands-on experience while delivering meaningful value to the organization.

EDUCATIONAL QUALIFICATION

- **Bachelor of Engineering – Artificial Intelligence and Machine Learning**
CMR Institute of Technology, Bengaluru
CGPA – 7.86, 2027
- **XII Grade CBSE - PCM**
Gyan Bharti Public School, Gaya
69% , 2022
- **X Grade CBSE**
Creane Memorial High School, Gaya
94% , 2020

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, SQL
- **Backend Frameworks:** FastAPI, Node.js
- **Frontend:** React.js, Tailwind CSS, HTML, CSS
- **AI/ Machine Learning:** Scikit-learn, Pandas, Numpy, CrewAI, Streamlit
- **Operating System:** Windows, Linux
- **Database & Caching:** MySQL, PostgreSQL , MongoDB, Redis
- **Messaging & Communications:** RabbitMQ, REST APIs, WebSockets
- **Security:** JWT(RS256), SPIFFE/SPIRE, Open Policy Agent(OPA), bcrypt
- **Tools & Infrastructure:** Git, RESTful APIs, Docker, Vercel, Kubernetes, Gradle, Vite, Alembic
- **Cloud Tools:** AWS(EC2, S3, Lambda), MS Azure

PROJECTS

- **YOLO-Based Smart Parking Detection System** [github/yolo](https://github.com/yolo)
 - Developed a real-time computer vision system using **YOLO** and **OpenCV** to detect vehicles and classify parking slot occupancy from images and video streams, enabling automated parking monitoring.
 - Trained and fine-tuned a custom **YOLO** object detection model on annotated parking datasets using transfer learning, applying data augmentation and hyperparameter optimization to improve detection performance.
 - Built an end-to-end inference pipeline supporting image, video, and live camera inputs with bounding box visualization, confidence scoring, and automated occupancy status generation for real-time monitoring.
 - Evaluated model performance using industry-standard object detection metrics including **Precision, Recall, mAP, and IoU**, while optimizing inference speed for deployment on resource-constrained systems.
 - **Tech Stack:** Python, YOLO, OpenCV, PyTorch, NumPy, Pandas, ONNX, Streamlit, Computer Vision, Object Detection.
- **Agentic AI Maritime Control Tower** [github/maritime](https://github.com/maritime)
 - Built an end-to-end multi-agent AI system that ingests maritime data (AIS vessel data, weather, port stats, geopolitical signals) to detect disruptions and generate decisions.
 - Architected a multi-agent orchestration pipeline with ML risk scoring for real-time vessel anomaly detection and disruption triage.
 - Developed a digital twin simulation to model sea routes and evaluate rerouting strategies, optimizing cost, delay, and disruption risk.

- Implemented LLM-based decision and explanation agents with human-in-the-loop feedback, deployed via a Streamlit dashboard for real-time, interpretable decision support.
 - **Tech Stack:** Python, CrewAI , Fast API, Streamlit, PostgreSQL, Pandas, NumPy , NetworkX, Plotly, REST APIs, Docker, Git, GitHub Actions
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- **Agent Identity & Access Management Platform** [github/agent-iam](https://github.com/agent-iam)
 - Built a production-grade IAM platform for AI agents with SPIFFE/SPIRE workload identity, RS256 JWT issuance (15-min TTLs), and zero-downtime API key rotation using bcrypt-hashed credential storage on a FastAPI/PostgreSQL backend.
 - Engineered a tamper-evident append-only audit log with SHA-256 hash chaining and causal trace propagation, linking every credential use, tool call, and access decision into a fully reconstructible event tree.
 - Implemented a multi-hop agent delegation engine with scope attenuation, cycle detection, and depth limiting — preventing privilege escalation across agent chains enforced via Open Policy Agent Rego policies.
 - Built an MCP tool-call proxy with pre-execution policy enforcement, SHA-256 argument/result hashing, and a buffered async audit flusher with exponential-backoff retry and dead-letter recovery.
 - **Tech Stack:** Python, FastAPI, PostgreSQL, SQLAlchemy (async), SPIFFE/SPIRE, Open Policy Agent, JWT (RS256), bcrypt, Docker, Alembic, asyncpg, httpx.
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- **AI-Powered ESG Performance Analytics Platform** [github/ESG](https://github.com/ESG)
 - Developed a full-stack AI-powered ESG analytics platform that analyzes Environmental, Social, and Governance metrics to predict organizational ESG risk levels and support sustainability-driven decision making.
 - Built machine learning pipelines using Random Forest and Gradient Boosting models to classify ESG risk, provide confidence scores, and generate ensemble-based predictions from company sustainability indicators.
 - Implemented RESTful APIs with FastAPI to process ESG datasets, perform real-time risk predictions, and deliver interactive analytics through a React-based dashboard with automated data visualization.
 - Designed an end-to-end data processing workflow including ESG data preprocessing, feature engineering, model training, and automated inference to enable scalable sustainability assessment and reporting.
 - Built an interactive dashboard with dynamic visualizations to present ESG analysis and key performance indicators.
 - **Tech Stack:** Python, FastAPI, Scikit-learn, Pandas, NumPy, Random Forest, Gradient Boosting, React.js, Vite, REST APIs, Docker, Pickle, HTML, CSS.

CERTIFICATES

- Full Stack Web Developer
- DSA in Python
- Complete Python Bootcamp
- Coursera: Foundations of CyberSecurity

LANGUAGE

English, Hindi Intermediate